higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 3

(6050403)

20 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use a pen with only BLUE or BLACK ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Which of the following meters is NOT an example of a positive displacement type?

A Rotating vane meter
B Piston meter
C Dry-gas bellows meter
D Target meter

1.2 Which of the following elements develops differential pressure proportional to flow?

A Orifice plate
B Pitot tube
C Ventral tube
D All the above-mentioned

1.3 What happens to fluid when it is forced to flow in a widening pipe?

A Flow becomes turbulent
B Pressure decreases
C Velocity decreases
D Volume decreases

1.4 Which ONE of the following statements about a conventional rotameter is true?

A Scale is non-linear
B Pressure drop is variable
C Metering is constant
D Design is self-cleaning

1.5 A/An … is the most common device used to restrict fluid flow for measuring.

A manometer
B orifice plate
C Venturi tube
D flow nozzle

(5 × 2) **[10]**

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QUESTION 2

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–I) next to the question number (2.1–2.8) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Manipulated space variable	A	only follows pre-arranged pattern feedback
2.2	Automatic control system	B	defines the set point
2.3	Linear feedback system	C	consists of two process variables
2.4	Controlled variable	D	an automatic control system in which different control sub-processes are connected one after the other
2.5	Manual control process	E	controls each step to make sure that they take place in the correct order
2.6	Open-loop control systems	F	speeds up a process by lowering disturbances and changing set points
2.7	Sequential operation	G	operators make changes in real time
2.8	Cascade control	H	the variable that the process tries to keep at the desired point
		I	control loop regulating a variable at a set point using algorithms and actuators

(8 × 2) **[16]**

TOTAL SECTION A: 26

SECTION B

QUESTION 3

3.1 Name the THREE group pressure instruments can be divided into. (3 × 2) (6)

3.2 Besides indicating the level of liquid, state TWO other things that can be indicated by sight glass. (2)

3.3 Briefly explain the use of a sight glass in measuring a level of liquid. (3)

3.4 Name FOUR types of mechanical gauges used in a process plant. (4)

3.5 Describe THREE scientific principles in which pressure can be measured electrically. (3)

3.6 Define *temperature*. (2)
[20]

QUESTION 4

4.1 Explain the following concepts:

4.1.1 Lower control limit

4.1.2 Normal control limit

4.1.3 Higher control limit

(3 × 2) (6)

4.2 Name FOUR basic instruments used in a process plant. (4)

4.3 Differentiate between *system error* and *parallax error*. (2 × 2) (4)

4.4 Explain a distributed control system which is also called a DCS. (2)
[16]

QUESTION 5

5.1 Determine the purpose of the following:

5.1.1 Indicator

5.1.2 Recorder

5.1.3 Controller

(3 × 2) (6)

5.2 Give the purpose of a legend in a process flow diagram (PFD). (2)

5.3 Draw the symbols of the following:

5.3.1 Main process

5.3.2 Insulated pipe

5.3.3 Electrically heated pipe

5.3.4 Pneumatic line

5.3.5 Hydraulic line

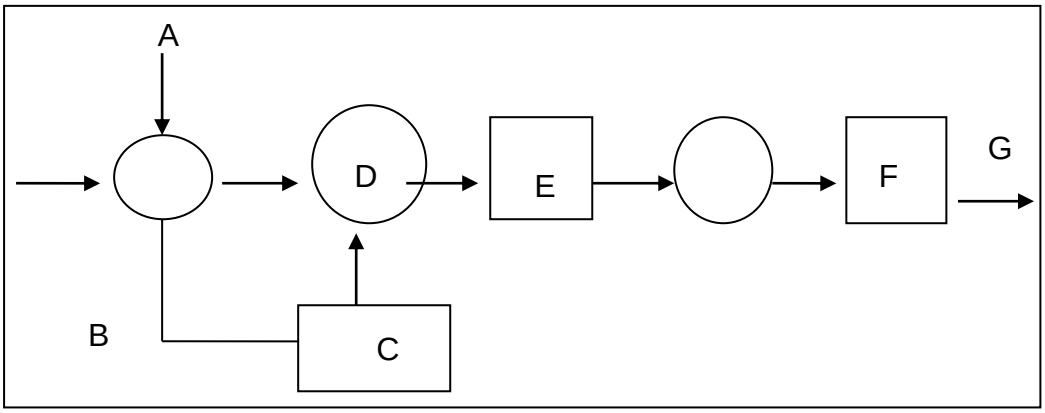
(5 × 2) (10)
[18]

QUESTION 6

6.1 Describe the purpose of *control loops*. (6)

6.2 State the TWO most common blending methods. (2)

6.3 Fill in the components of the following feed-forward control mechanism:



(7)
[15]

QUESTION 7

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (7.1–7.5) in the ANSWER BOOK.

7.1 The main function of an instrument is to measure the value of a process variable.

7.2 Throughput indicates the quantity of measurable items you can pass through the system.

7.3 An elastic element goes back to its original shape after it has been deframed.

7.4 A fail-safe system is designed to keep the power on.

7.5 Interlocks are used where personal safety is important.

(5 × 1) **[5]**

TOTAL SECTION B: 74
GRAND TOTAL: 100

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